

Cloud Architecture Design in Microsoft Azure

Case Study: E-Commerce Platform Migration and Scalability

Level: Fundamentals & Application

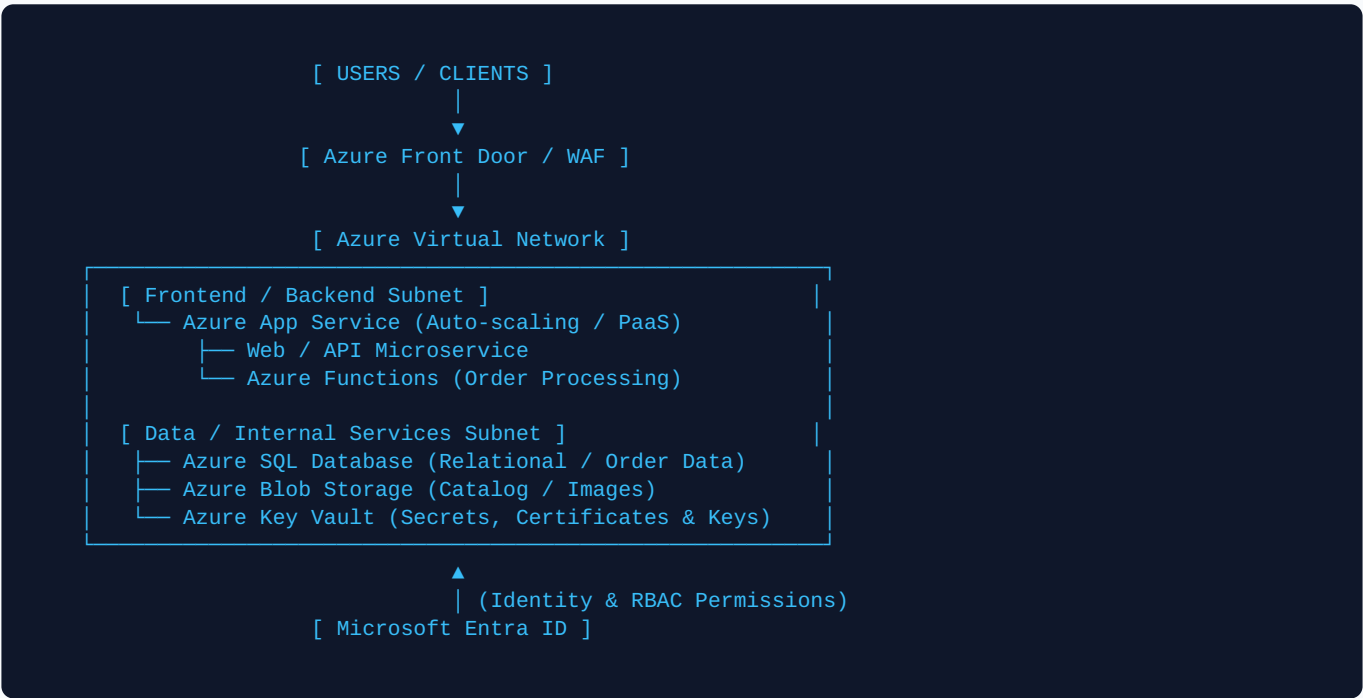
Aligned with AZ-900

Microsoft Azure

1. Project Executive Summary

This document outlines the cloud infrastructure architecture designed in **Microsoft Azure** to migrate a traditional e-commerce platform. The primary objective is to eliminate infrastructure bottlenecks during high-demand seasons (traffic peaks), maximize security in handling banking transactions, and optimize operational costs using managed services (PaaS).

2. Solution Architecture Diagram



3. Component Matrix and Technical Justification

Azure Service	Model	Role in Architecture	Selection Justification
Azure App Service	PaaS	Web & API Hosting	Eliminates OS maintenance. Automatic vertical and horizontal scaling in minutes.
Azure Functions	PaaS / Serverless	Asynchronous Processing	Executes event-driven code (confirmation emails, invoices). Strict pay-per-use billing.
Azure SQL Database	PaaS	Transactional Database	Built-in high availability (99.99%), automated backups, and data encryption at rest.
Azure Blob Storage	IaaS/PaaS	Unstructured Storage	Massive storage for product images. Hot/Cool tiers for cost optimization.
	SaaS		

Azure Service	Model	Role in Architecture	Selection Justification
Microsoft Entra ID		Identity & Access Management	Role-Based Access Control (RBAC) and MFA for technical and administrative staff.
Azure Key Vault	PaaS	Secrets Management	Centralization and encryption of DB passwords, API tokens, and SSL certificates.

4. Security, Governance, and Cloud FinOps

Defense-in-Depth Security Strategy

Network isolation using private Subnets within Azure Virtual Network (VNet). Malicious traffic filtering with Web Application Firewall (WAF) and principle of least privilege using RBAC.

Financial Estimation and Cost Governance (FinOps)

Using the **Azure Pricing Calculator**, a baseline environment is projected with an estimated operational cost of **~180 USD/month** in the *West Europe* region. To maintain economic sustainability, the following policies are implemented:

- **Reserved Instances:** 1 or 3-year commitment for Azure SQL Database to obtain discounts of up to 45%.
- **Auto-scaling Rules:** Automatic reduction of App Service instances during low-demand night hours.
- **Budgets and Alerts:** Alert configuration in *Azure Cost Management* when reaching 80% of the estimated monthly budget.